Ames Housing Data Analysis ML Foundations Bootcamp — Capstone Project

Exploring what drives house prices in Ames, Iowa
Dataset: 2,930 sales | 80+ features | Python & Jupyter

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Executive Summary

Introduction This project analyses the Ames Housing dataset, containing 2,930 house sales in Ames, Iowa with over 80 features covering lot size, quality, neighbourhood, and sale price. Questions we aimed to answer: - What features most strongly predict sale price? - Does house age and quality affect price? - Can engineered features outperform raw data?

Final shape: 2,929 rows × 78 columns

Cleaning Summary

1 — Wrong Data Types
MS SubClass and Mo Sold converted from int to str and category types.

2 — Missing Values
- Dropped 4 columns with > 80% missing
- Filled garage/basement columns with 'None' or 0
- Filled Lot Frontage with median (17% missing)

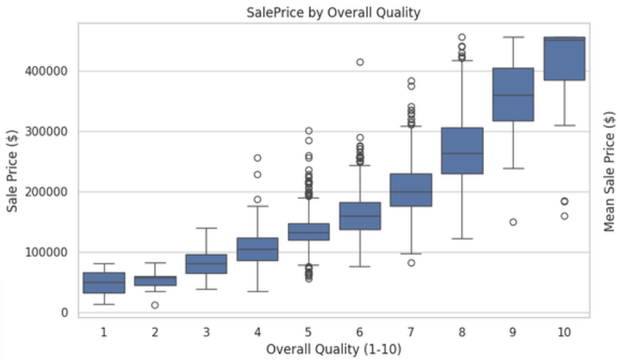
3 — Outliers
137 extreme values capped at $456,690 (99th pct)

4 — Duplicates
0 duplicate rows found.

Key Findings

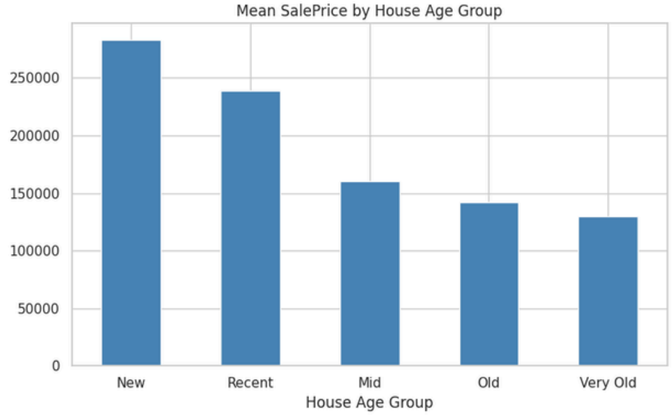
1 — Quality is the strongest price driver

Houses rated quality 10 have a median price nearly 10x higher than quality 1 houses. Correlation with SalePrice = 0.81



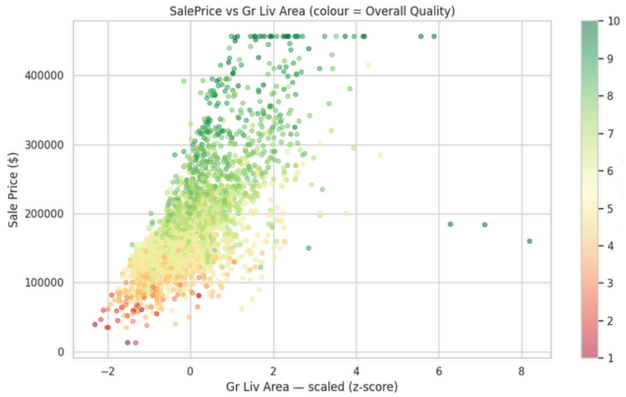
2 — New houses cost 2.2x more than very old

New = $283k average vs Very Old = $129k. House age is a consistent price predictor.



3 — Size and quality jointly drive price (Best Chart)

High-quality houses (green) consistently appear at the top of the price range for any given size. Low-quality houses (red) cluster at the bottom. Neither size nor quality alone tells the full story.



Feature Engineering Summary

Feature Engineering Summary 8 new features created: - One-hot encoding: MS Zoning, Neighborhood (+31 cols) - Ordinal encoding: Kitchen Qual → 1 to 5 - Scaled: Gr Liv Area, Lot Area (mean=0, std=1) - price_per_sqft: cost per unit area - bath_per_bed: comfort ratio - qual_x_area: strongest predictor (r = 0.84) - log1p(Lot Area): removed right skew - House Age Group: 5 meaningful bins

What I Would Do Next

01

Build a predictive model using Random Forest or Gradient Boosting on the engineered features.

02

Neighbourhood deep-dive — explore whether location explains price beyond quality.

03

Seasonal analysis — does month of sale affect price?

04

Feature selection using Lasso regression to find the most impactful features.